

type of shock is shown with an asterisk.

In the previous session, we discussed preload, afterload, cardiac output (CO), and mean arterial pressure (MAP) as the primary physiologic determinants of perfusion. In clinical practice, particularly in critically ill patients, additional hemodynamic measurements help us better classify shock and guide therapy. These include **right atrial pressure (RAP)**, **pulmonary capillary wedge pressure (PCWP)**, **cardiac index (CI)**, and **mixed venous oxygen saturation (SvO₂)**. Together, these measurements help localize where in the circulatory system the problem is occurring — venous return, ventricular filling, pump function, or tissue oxygen utilization.

	Mean arterial blood pressure (MAP)	End-diastolic volume (preload)	Systemic vascular resistance	Cardiac output	Heart rate
Distributive	↓	↓	↓P	↑	↑
Cardiogenic	↓	↑	↑	↓P	↑
Hypovolemic	↓	↓P	↑	↓	↑
Neurogenic	↓P	↓P	↓P	↓P	*Normal or ↓
Obstructive: pulmonary	↓	*↑ (RV) ↓ (LV)	↑	↓P	↑
Obstructive: mechanical	↓	↓	↑	↓P	↑

Right Atrial Pressure (RAP): Right atrial pressure reflects the pressure in the

CLINICAL REASONING

PCWP, like RAP helps differentiate cardiogenic and obstructive shock from hypovolemic shock.

Cardiac Index (CI) Cardiac index adjusts cardiac output for body surface area

CI = CO/BSA

Normal CI is approximately **2.5–4.0 L/min/m²**. A reduced CI indicates inadequate forward flow relative to patient size and is seen in cardiogenic, hypovolemic, and obstructive shock. In early distributive (septic) shock, CI may be normal or elevated due to decreased systemic vascular resistance and compensatory hyperdynamic circulation. (hyperdynamic = decrease in SVR due to endotoxin release leads to increase in CO, however, in late septic shock SVR may increase)

Mixed Venous Oxygen Saturation (SvO₂) Mixed venous oxygen saturation reflects the percentage of oxygen remaining in blood returning to the heart after tissue extraction. It represents the balance between oxygen delivery (DO₂) and oxygen consumption (VO₂). Normal SvO₂ ranges from **60–80%**. Low SvO₂ indicates that tissues are extracting more oxygen because delivery is inadequate (low cardiac output states such as cardiogenic or hypovolemic shock). Elevated SvO₂ may be seen in early septic shock due to impaired oxygen extraction at the tissue level.

Placement of a **right** can determine factored right ventricle or normal (same), unless the patient has right ventricular pulmonary edema (more on that in a minute).

Administering IV fluids is the most effective treatment of **hypovolemic shock**. Here, the primary insult is low cardiac preload. The lower preload causes reductions in stroke volume and cardiac output, which, in turn, cause low blood pressure. Fluid therapy increases preload, raising cardiac output. The effect is most marked in those with hypovolemic shock but may be useful as well in those with mild distributive shock. Blood can be used instead of IV fluids but is generally reserved for those with overt hemorrhage.

Giving IV fluids carries little risk, which is why IV fluids are used in most hypotensive patients, even before the precise cause of low blood pressure or shock is identified. In fact, the patient's response to saline is often diagnostic: if they respond promptly, you are probably dealing with hypovolemic shock and not one of the other types.

When should you **not** give IV fluids? Patients should not receive IV fluids if they have pulmonary edema, diagnosed hypoxia, and interstitial edema on chest x-ray ([Figure 1](#)).

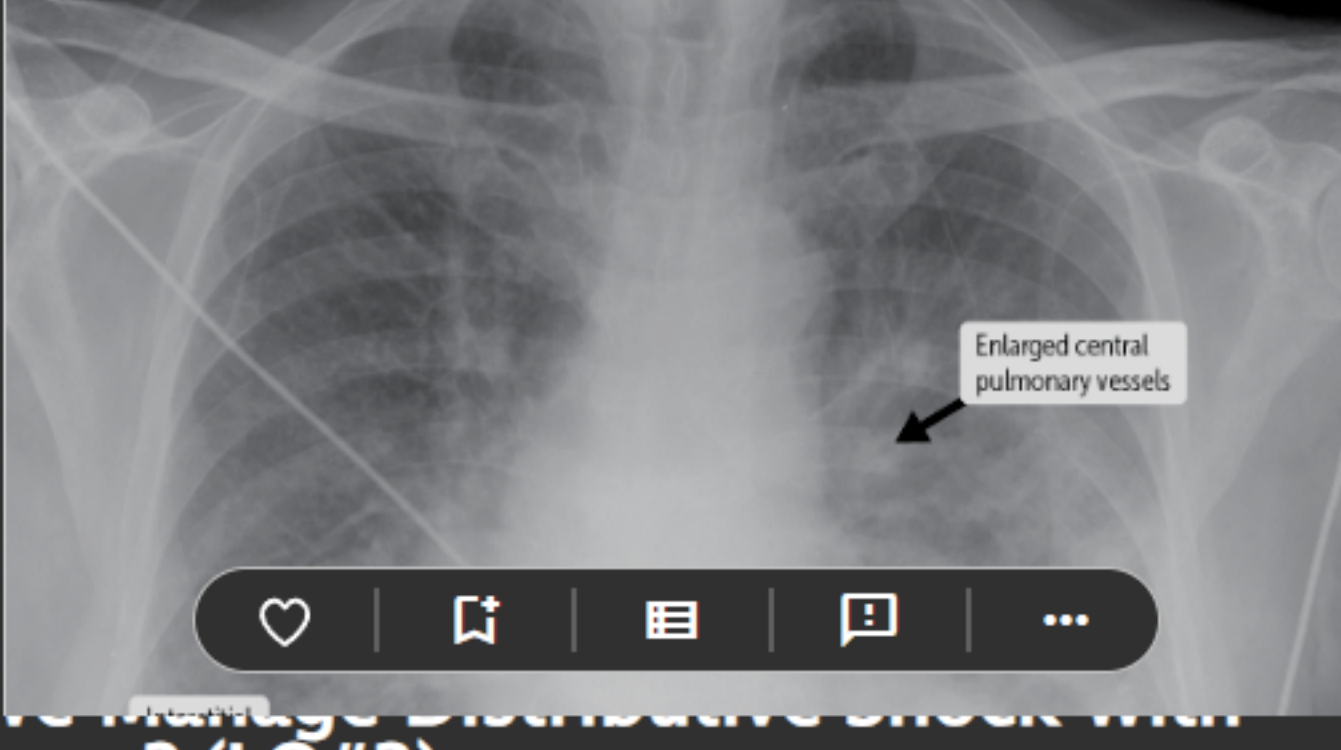


Figure 1. Chest x-ray showing interstitial edema and pulmonary vascular congestion.

Vasopressors? (LO#3)

As you see in Table 1, the different forms of shock have different physiologic patterns. Many of the therapies for shock are directed specifically against one of these abnormalities. We will now work through some of these, beginning with reduced systemic vascular resistance (SVR), one of the characteristics of distributive shock.

Distributive shock is caused by arteriolar vasodilation, which reduces SVR. Therefore, **vasopressor drugs** (sometimes called “pressors”), which cause constriction of the arterioles, are an effective treatment. The following is a list of receptors and the pressor drugs that activate them to achieve vasoconstriction:

- α₁-Adrenergic, activated by norepinephrine, epinephrine, or phenylephrine
- Vasopressin V_{1A}, activated by vasopressin
- Dopaminergic, activated by dopamine

Norepinephrine

Although the use of these pressor drugs varies and can be customized for an individual patient, norepinephrine is the drug of choice for most patients with distributive shock, as has been shown in survival studies. It acts primarily at the α₁ adrenergic receptor, secondarily at the β₁ and β₂ receptors. The drug causes vasoconstriction and tachycardia (via β₁ stimulation).

An adverse effect is tachyarrhythmias, although the risk is lower than that of stronger β₁ agonists. Like the other pressor drugs, norepinephrine should only be used when absolutely necessary. the drugs may

is to induce vasodilation in the renal circulation. The degree of activity on each type of receptor is dose-dependent: low doses are primarily dopaminergic, and increasing doses produce inotropic effects via β₁ stimulation, then systemic vasoconstriction α₁ stimulation. Side effects include tachycardia and tachyarrhythmias, as is the case with most drugs with inotropic effects.

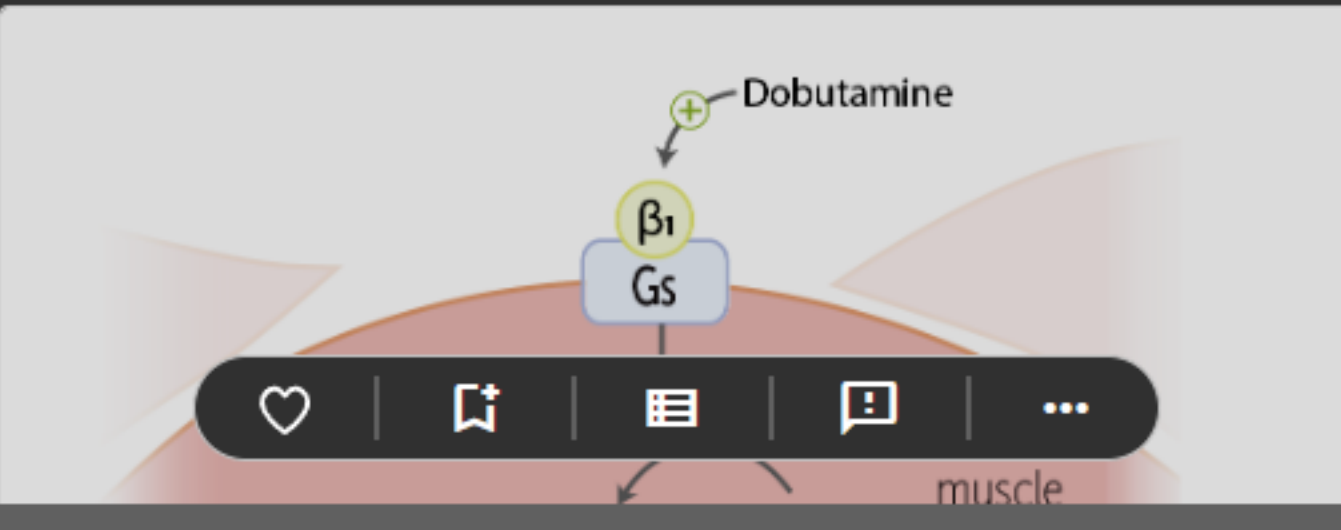
CLINICAL CORRELATION

Because acute kidney injury is so common in shock-states, the theoretical benefit of low-dose dopamine is to preserve blood flow to the kidneys that sustains renal function. However, there has been no proven renal-protective benefit in the clinical setting, and increasing renal blood flow may lead to diuretic effect that can cause significant harm in a hypotensive patient.

These drugs can be used in combination, and it is common to mix drugs with different receptor targets (eg, use norepinephrine with vasopressin). Table 2 summarizes the effects of the vasopressor drugs on the various receptors; the number of plus signs indicates relative receptor affinity. (This is a must know table !!)

Table 2 Vasopressor receptor targets					
Drug	α ₁	β ₁	β ₂	V _{1A}	Dopaminergic
Norepinephrine					None

- Dobutamine;Dobutamine strongly stimulates β₁ receptors, causing an **increase in cardiac contractility and heart rate** and, therefore, cardiac output. By weakly stimulating β₂ receptors, dobutamine also causes mild arteriolar vasodilation, reducing SVR and afterload; hence, it is often helpful in treating heart failure. It has minimal effect on α₁ receptors, so causes little or no vasoconstriction. Dobutamine can raise the heart rate substantially and can also precipitate ventricular and atrial arrhythmias through its β₁ stimulation. Because it causes vasodilation, it can sometimes lower blood pressure.
- The inotrope **milrinone** works differently; it inhibits phosphodiesterase 3 (PDE3) see Figure 2). PDE3 lies downstream, inside smooth muscle cells and on the β₁ pathway; PDE3 degrades cyclic adenosine monophosphate (cAMP) into AMP, which leads to a decrease in intracellular Ca²⁺ and, therefore, a decrease in contractility. By inhibiting PDE3, milrinone decreases cAMP degradation and thus increases myocyte intracellular Ca²⁺ and cardiac contractility. This has the effect of **increasing stroke volume and cardiac output**. By inhibiting PDE3, milrinone also causes vasodilation throughout the vasculature. Milrinone's combined effect of increasing contractility and decreasing afterload makes it great for treating cardiogenic shock.



Mechanical Devices

Some patients with severe cardiogenic shock do not respond to drugs alone and require placement of a **mechanical assist device** to support them until their hearts recover. None of the devices described below have demonstrated clear improvements in patient outcomes, including mortality. These devices should not be used in patients with sepsis, bleeding, or severe aortic disease:

- **Intra-aortic balloon pump (IABP)**: This is a catheter-inserted pump that lies in the proximal aorta and coordinates with the ECG to assist the heart in pumping blood forward ([Figure 3](#)).
- **Percutaneous left ventricle assist devices**: These require surgical placement and come with a higher risk than IABP. They can be used to support the left ventricle, even when a thrombus is present in-situ, and can also provide support if right ventricular failure occurs. In addition, these devices can be used in the presence of a prosthetic aortic valve or aortic stenosis.
- **Extracorporeal membrane oxygenation (ECMO)**: This provides complete cardiopulmonary support for patients with both cardiac and respiratory failure. It involves pumping the patient's blood into a heart-lung machine for oxygenation and then returning it to the patient. Full hemodynamic support can be provided by this method. There are currently distributive shock (A) arises from systemic vasodilation, often due to sepsis or anaphylaxis. Cardiogenic shock (B) results from heart contractions leading to low cardiac output. Hypovolemic shock (C) is due to low intravascular volume. Obstructive shock (E) involves blood flow obstruction.

How would you rate this Brick?

References

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Congenital Heart Defects: Foundations and Frameworks

Last updated March 10, 2026

20 min read

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Learning Objectives (2)

Versions

After completing this brick, you will be able to:

1

Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.

2

Create a congenital heart defect (CHD) differential diagnosis, and pathophysiology for congenital